LIMITS

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Name: Solutions

In each of the problems, evaluate the limit if it exists. Indicate any limit laws that you use. If the limit does not exist, explain why.

1. Use the limits

$$\lim_{x\to 2} f(x) = 4 \qquad \lim_{x\to 2} g(x) = -2 \qquad \lim_{x\to 2} h(x) = 0$$

to complete each of the following.

(a) $\lim_{x\to 2}[f(x) + 5g(x)]$

$$\overset{1}{=} \lim_{x\to 2} f(x) + \lim_{x\to 2} 5g(x) \overset{3}{=} 4 + 5\lim_{x\to 2} g(x) = 4 + 5(-2) = 4 - 10 = \boxed{-6.}$$

(b) $\lim_{x\to 2} [g(x)]^3$

$$= \lim_{x\to 2} g(x)\cdot g(x)\cdot g(x) \overset{4}{=} \lim_{x\to 2} g(x) \lim_{x\to 2} g(x)g(x)$$

$$\overset{4}{=} \lim_{x\to 2} g(x) \lim_{x\to 2} g(x) \lim_{x\to 2} g(x) = \left(\lim_{x\to 2} g(x)\right)^3 = (-2)^3 = \boxed{-8.}$$

(c) $\lim_{x\to 2} \sqrt{f(x)}$

$$= \sqrt{\lim_{x\to 2} f(x)} = \sqrt{4} = \boxed{2.}$$

(d) $\lim_{x\to 2} \frac{3f(x)}{g(x)}$

$$\overset{⑥}{=} \frac{\lim_{x\to 2} 3f(x)}{\lim_{x\to 2} g(x)} \overset{③}{=} \frac{3\lim_{x\to 2} f(x)}{-2} = \frac{3(4)}{-2} = 3(-2) = \boxed{-6.}$$

(e) $\lim_{x\to 2} \frac{g(x)}{h(x)}$ $\boxed{\text{Does Not Exist.}}$

If $h(x)<0$ near $x=2$, then $\frac{g(x)}{h(x)} \to \infty$

If $h(x)>0$ near $x=2$, then $g(x)/h(x) \to -\infty$

If $h(x)$ switches sign around $x=2$, then the left and right limits are opposite infinities: either $\lim_{x\to 2^+} \frac{g(x)}{h(x)} = \infty$ and $\lim_{x\to 2^-} \frac{g(x)}{h(x)} = -\infty$ or

$\lim_{x\to 2^+} \frac{g(x)}{h(x)} = -\infty$ and $\lim_{x\to 2^-} \frac{g(x)}{h(x)} = \infty$

(f) $\lim_{x\to 2} \frac{g(x)h(x)}{f(x)}$

$$\overset{⑤}{=} \frac{\lim_{x\to 2} g(x)h(x)}{\lim_{x\to 2} f(x)} = \frac{\lim_{x\to 2} g(x) \lim_{x\to 2} h(x)}{4} = \frac{(-2)(0)}{4} = \frac{0}{4} = \boxed{0.}$$

2. $\lim_{x\to -1} (x^4 - 3x)(x^2 + 5x + 3)$

$$\overset{④}{=} \lim_{x\to -1} (x^4-3x) \lim_{x\to -1} (x^2+5x+3) = \left((-1)^4 - 3(-1)\right)\left((-1)^2 + 5(-1) + 3\right)$$

$$= (1+3)(1-5+3)$$

$$= 4(-1) = \boxed{-4.}$$

3. $\lim_{u\to -2} \sqrt{u^4 + 3u + 6}$

$$= \sqrt{\lim_{u\to -2} u^4+3u+6} = \sqrt{(-2)^4+3(-2)+6} = \sqrt{16-6+6} = \sqrt{16} = \boxed{4.}$$

4. $\lim_{t\to 2} \left(\frac{t^2-2}{t^3-3t+5}\right)^2$

$$= \left(\lim_{t\to 2} \frac{t^2-2}{t^3-3t+5}\right) \overset{⑤}{=} \left(\frac{\lim_{t\to 2} t^2-2}{\lim_{t\to 2} t^3-3t+5}\right)^2 = \left(\frac{2^2-2}{2^3-3(2)+5}\right)^2$$

$$= \left(\frac{4-2}{8-6+5}\right)^2 = \left(\frac{2}{7}\right)^2 = \boxed{\frac{4}{49}.}$$

5. $\lim_{t\to -3} \frac{t^2-9}{2t^2+7t+3}$

$2t^2+7t+3 = (2t+1)(t+3)$

$$\lim_{t\to -3} \frac{t^2-9}{2t^2+7t+3} = \lim_{t\to -3} \frac{(t+3)(t-3)}{(t+3)(2t+1)} = \lim_{t\to -3} \frac{t-3}{2t+1}$$

$$\overset{5}{=} \frac{\lim_{t\to -3} t-3}{\lim_{t\to -3} 2t+1} = \frac{-3-3}{2(-3)+1} = \frac{-6}{-6+1} = \boxed{\frac{6}{5}}$$

6. $\lim_{x\to -3} \frac{x^2+3x}{x^2-x-12}$

$x^2-x-12 = (x-4)(x+3)$

$$\lim_{x\to -3} \frac{x(x+3)}{(x-4)(x+3)} = \lim_{x\to -3} \frac{x}{x-4} = \frac{-3}{-3-4} = \frac{-3}{-7} = \boxed{\frac{3}{7}}$$

7. $\lim_{x\to -2} \frac{x+2}{x^3+8}$

[Hint: $x^3+a^3 = (x+a)(x^2-ax+a^2)$]

$$\lim_{x\to -2} \frac{x+2}{x^3+8} = \lim_{x\to -2} \frac{x+2}{(x+2)(x^2-2x+4)} = \lim_{x\to -2} \frac{1}{x^2-2x+4} = \frac{1}{(-2)^2-2(-2)+4} = \frac{1}{4+4+4} = \boxed{\frac{1}{12}}$$

8. $\lim_{h\to 0} \frac{(2+h)^3-8}{h}$

$$= \lim_{h\to 0} \frac{2^3+3(2^2)(h)+h^2-8}{h} = \lim_{h\to 0} \frac{8+12h+h^2-8}{h} = \lim_{h\to 0} 12+h = \boxed{12}$$

9. $\lim_{t\to 0} \left(\frac{1}{t} - \frac{1}{t^2+t}\right)$

$$= \lim_{t\to 0} \left(\frac{t+1}{t(t+1)} - \frac{1}{t(t+1)}\right) = \lim_{t\to 0} \frac{t}{t(t+1)} = \lim_{t\to 0} \frac{1}{t+1} = \boxed{1}$$

10. $\lim_{t\to 0}\left(\frac{\sqrt{1+t}-\sqrt{1-t}}{t}\right)$

$$= \lim_{t\to 0} \frac{\sqrt{1+t}-\sqrt{1-t}}{t} \cdot \frac{\left(\sqrt{1+t}+\sqrt{1-t}\right)}{\left(\sqrt{1+t}+\sqrt{1-t}\right)} = \lim_{t\to 0} \frac{(1+t)-(1-t)}{t\left(\sqrt{1+t}+\sqrt{1-t}\right)}$$

$$= \lim_{t\to 0} \frac{2t}{t\left(\sqrt{1+t}+\sqrt{1-t}\right)} = \lim_{t\to 0} \frac{2}{\sqrt{1+t}+\sqrt{1-t}} = \frac{2}{\sqrt{1+0}+\sqrt{1-0}} = \frac{2}{1+1} = \frac{2}{2} = \boxed{1.}$$

11. $\lim_{h\to 0} \frac{\frac{1}{(x+h)^2}-\frac{1}{x^2}}{h}$

$$= \lim_{h\to 0} \frac{1}{h}\left(\frac{x^2}{x^2(x+h)^2} - \frac{(x+h)^2}{x^2(x+h)^2}\right) = \lim_{h\to 0} \frac{1}{h}\left(\frac{x^2-x^2-2xh-h^2}{x^2(x+h)^2}\right)$$

$$= \lim_{h\to 0} \frac{-2x-h}{x^2(x+h)^2} = \frac{-2x-0}{x^2(x+0)^2} = \frac{-2x}{x^4} = \boxed{-\frac{2}{x^3}.}$$